

Unit I
Linear Programming

Course Objective – COB1: To enable students understand linear programming problems.

Course Outcome – CO1: To solve linear programming problems through simplex and other methods.

Bloom's Taxonomy Level: K1, K2, K3, K4

Part A

S.No.	Question	BTL*
1	What are the characteristics of LPP?	K1
2	Write the standard form of LPP.	K2
3	Define feasible solution	K1
4	Define optimum solution.	K1
5	What is unbounded solution?	K1
6	What is the use of introducing slack and surplus variables?	K1
7	Distinguish between the graphical and simplex methods of solving LP problems.	K2
8	Define alternate optimum solution.	K1
9	Define slack variable in an LPP.	K1
10	Define surplus variable in an LPP.	K1
11	Define LPP.	K1
12	What do you mean by decision variables?	K1
13	State the advantages of the simplex method over the graphical method	K2
14	What is basic variable and non-basic variable?	K1
15	What is meant by a pivot element?	K1
16	List any two applications of LPP in real life..	K2
17	What is an artificial variable?	K2
18	What are the limitations of LPP?	K1
19	Distinguish between Maximization and Minimization problems.	K2
20	When do we use artificial variable in Big-M Method.	K1

Part B

S.No.	Question	BTL *
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	(i) A company has 3 operational departments namely weaving, processing & packing, with capacity to produce three different types of clothes namely suiting, shirting & woollens yielding a profit of Rs.2, Rs.4 & Rs.3 per meter. One meter of suiting requires 3 minutes of weaving, 2 minutes of processing & 1 minute of packing. One meter of woollen requires 3 minutes in each department. One meter of shirt required 4 minutes, 1 minute & 3 minutes in weaving, process & packing respectively. In a week, total run time of each department is 60, 40 & 80 hours. Formulate as an LPP. Can this be solved by Graphical or Simplex method? If yes, Justify it. (16)																			
1	(ii) A person wants to decide the constituents of a diet which will fulfil his daily requirements of proteins, fat and carbohydrates at the minimum cost. The choice is to be made from four different types of foods. A diet for the person must contain minimum 800 units of proteins, 200 units of fats and 300 units of carbohydrates. Four types of food are available at a cost of Rs.45, Rs.40, Rs.85, & Rs.65 per unit respectively. Type-1 food contains 3 units of proteins, 2 unit of fats & 6 carbohydrates, Type-2 food contains 4 units of proteins, 2 unit of fats & 4 carbohydrates, Type-3 food contains 8 units of proteins, 7 unit of fats & 7 carbohydrate and Type-4 food contains 6 units of proteins, 5 unit of fats & 4 carbohydrates. Formulate LPP for the given situation. Can this be solved by Graphical or Simplex method? If yes, Justify it. (16)	K4																		
	(i) A diet for a sick person must contain at least 4,000 units of vitamins, 50 units of minerals, and 1,400 calories. Two foods A & B are available at a cost of Rs.4 & Rs.3 per unit respectively. If one of A contains 200 units of vitamins, 1 unit of mineral & 40 calories, and one unit of food B contains 100 units of vitamins, 2 units of minerals & 40 calories, formulate the problem as an LP model and solve it by graphical method to find a combination of foods for the least cost. (16) (ii) A company manufactures two types of products P_1 and P_2. Each product uses lathe and milling machine. The processing time per unit of P_1 on the lathe is 5 hours and on the milling machine is 4 hours. The processing time per unit of P_2 on the lathe is 10 hours and on the milling machine, 4 hours. The maximum number of hours available per week on the lathe and the milling machine are 60 hours and 40 hours, respectively. Also the profit per unit of selling P_1 and P_2 are Rs 6.00 and Rs 8.00 respectively. Formulate as an LP model and solve it graphically. (16) (iii) A nutrition scheme for babies is proposed by a committee of doctors. Babies can be given two types of food, I and II, which are available in standard sized packets weighing 50 grams. The cost per packet of these foods is Rs 2 and Rs 3, respectively. The vitamin availability in each type of food per packet and the minimum vitamin requirement for each type of vitamin are summarized in table. Develop a linear programming model and solve it graphically. (16) <table><tr><th rowspan="2">Vitamin</th><th colspan="2">Vitamin availability per packet</th><th rowspan="2">Minimum daily requirement</th></tr><tr><th>Food Type I</th><th>Food Type II</th></tr><tr><td>1</td><td>1</td><td>1</td><td>6</td></tr><tr><td>2</td><td>7</td><td>1</td><td>14</td></tr><tr><td>Cost/packet (Rs)</td><td>2</td><td>3</td><td></td></tr></table>	Vitamin	Vitamin availability per packet		Minimum daily requirement	Food Type I	Food Type II	1	1	1	6	2	7	1	14	Cost/packet (Rs)	2	3		K4
Vitamin	Vitamin availability per packet		Minimum daily requirement																	
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	iv. A small workshop makes Tables and Chairs. Each table requires 4 hours of carpentry and 3 hours of painting. Each chair requires 3 hours of carpentry and 1 hour of painting. The workshop has at most 24 hours of carpentry and 9 hours of painting available per week. Profit is ₹60 per table and ₹30 per chair. Formulate and solve using Graphical Method. (16)	
3	Solve the following LPP using graphical method. (i) Maximize $z = 3x_1 + 2x_2$ subject to, $-2x_1 + x_2 \leq 1$ $x_1 \leq 2$ $x_1 + x_2 \leq 3$ $x_1, x_2 \geq 0$ (16) (ii) Maximize $z = x_1 - 2x_2$ subject to, $-x_1 + x_2 \leq 1$ $6x_1 + 4x_2 \geq 24$ $0 \leq x_1 \leq 5 \quad \& \quad 2 \leq x_2 \leq 4$ (16) (iii) Minimize $z = 45x_1 + 55x_2$ subject to, $x_1 + 2x_2 \leq 30$ $2x_1 + 3x_2 \leq 80$ $x_1 - x_2 \geq 8$ $x_1, x_2 \geq 0$ (16) (iv) Maximize $z = 6x_1 + 8x_2$ subject to, $5x_1 + 10x_2 \leq 60$ $4x_1 + 4x_2 \leq 40$ $x_1, x_2 \geq 0$ (16)	K4
4	Use simplex method to solve the following LPP. (i) Minimize $z = 8x_1 - 2x_2$ subject to, $-4x_1 + 2x_2 \leq 1$ $5x_1 - 4x_2 \leq 3$ $x_1, x_2 \geq 0$ (16) (ii) Maximize $z = 10x_1 + 15x_2 + 20x_3$ subject to, $2x_1 + 4x_2 + 6x_3 \leq 24$ $3x_1 + 9x_2 + 6x_3 \leq 30$ $x_1, x_2, x_3 \geq 0$ (16) (iii) Maximize $z = 6x_1 + 8x_2$ subject to, $5x_1 + 10x_2 \leq 60$	K4

	$4x_1 + 4x_2 \leq 40$ $x_1, x_2 \geq 0$ (16) (iv) Maximize $z = 3x_1 + 5x_2$ subject to, $x_1 + 2x_2 \leq 8$ $3x_1 + 2x_2 \leq 12$ $x_1, x_2 \geq 0$ (16)	
5	Use Big M-method to solve the following LPP. (i) Minimize $z = 2x_1 + 3x_2$ subject to, $x_1 + x_2 \geq 6$ $7x_1 + x_2 \geq 14$ $x_1, x_2 \geq 0$ (16) (ii) Maximize $Z = 20x_1 + 10x_2 + 15x_3$ subject to the constraints, $8x_1 + 6x_2 + 2x_3 \leq 60$ $5x_1 + x_2 + 6x_3 \geq 40$ $2x_1 + 6x_2 + 3x_3 \leq 30$ $x_1, x_2, x_3 \geq 0$ (16) (iii) Minimize $z = 12x_1 + 18x_2 + 15x_3$ subject to, $4x_1 + 8x_2 + 6x_3 \geq 64$ $3x_1 + 6x_2 + 12x_3 \geq 96$ $x_1, x_2, x_3 \geq 0$ (16)	K4
6	Use two-phase method to solve the following LPP. (i) Minimize $z = 12x_1 + 18x_2 + 15x_3$ subject to, $4x_1 + 8x_2 + 6x_3 \geq 64$ $3x_1 + 6x_2 + 12x_3 \geq 96$ $x_1, x_2, x_3 \geq 0$ (16) (ii) Minimize $Z = x_1 + x_2$ subject to, $2x_1 + x_2 \geq 4$ $2x_1 + 7x_2 \geq 7$ $x_1, x_2 \geq 0$ (16) (iii) Maximize $Z = 20x_1 + 10x_2 + 15x_3$ subject to the constraints, $8x_1 + 6x_2 + 2x_3 \leq 60$ $5x_1 + x_2 + 6x_3 \geq 40$ $2x_1 + 6x_2 + 3x_3 \leq 30$ $x_1, x_2, x_3 \geq 0$ (16)	K4

